

PROJECT DESIGN PHASE

Proposed Solution

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S. No	Parameter	Description
1	Problem Statement (Problem to be solved)	Electricity consumption data is generated across multiple states and regions, but analyzing this large volume of data manually is difficult. Government departments and energy analysts often lack interactive tools to identify consumption trends, compare regional electricity usage, and support data-driven decision-making for efficient energy management.
2	Idea / Solution Description	The proposed solution uses an electricity consumption dataset stored in a MySQL database. SQL queries are used for data retrieval and analysis, while Tableau Desktop is used to build interactive dashboards and stories showing state-wise usage, monthly trends, region-wise comparison, top electricity consumers, and business insights. These dashboards are published on Tableau Public and embedded into a Flask web application, allowing users to analyze electricity consumption through an easy-to-use web interface.
3	Novelty / Uniqueness	Instead of traditional reports, the solution provides interactive dashboards with filters that allow users to explore electricity consumption data dynamically. Integrating MySQL, Tableau, Tableau Public, and Flask creates a complete analytics platform that improves visualization, accessibility, and decision-making for electricity management.
4	Social Impact / Customer Satisfaction	Electricity boards, government organizations, researchers, and analysts can better understand electricity demand patterns across different states and regions. This helps improve electricity distribution planning, optimize resource allocation, support sustainable energy management, and provide better services to consumers.
5	Business Model (Revenue Model)	The solution can be deployed as a subscription-based analytics platform for electricity boards, utility companies, and government organizations. Revenue can also be generated through customized dashboard development, enterprise analytics solutions, maintenance services, and consulting for energy management projects.

S. No	Parameter	Description
6	Scalability of the Solution	The solution can be expanded by integrating real-time electricity consumption data, smart meter information, IoT devices, AI-based electricity demand forecasting, renewable energy analytics, and support for district-wise and city-wise electricity monitoring without significant changes to the existing system architecture.

Workflow

The **Electricity Consumption Analysis** system follows a structured workflow to convert raw electricity consumption data into meaningful visual insights. Each stage of the workflow contributes to accurate data processing, interactive visualization, and effective decision-making.

- **Step 1: Data Collection**

The electricity consumption dataset is collected in **Excel (.xlsx)** format. The dataset contains information such as state, region, month, electricity consumption, and other related attributes.

- **Step 2: Data Preprocessing**

The collected data is cleaned by removing duplicate records, handling missing values, correcting inconsistencies, and validating the dataset to ensure accuracy before analysis.

- **Step 3: Database Management**

The cleaned dataset is imported into the **MySQL** database. SQL tables are created, and queries are executed to retrieve, filter, aggregate, and analyze electricity consumption data efficiently.

- **Step 4: Data Visualization**

The processed data is connected to **Tableau Desktop**, where multiple worksheets, dashboards, maps, and Tableau Stories are developed. Interactive charts provide state-wise analysis, region-wise comparison, monthly electricity trends, top electricity-consuming states, and business insights.

- **Step 5: Dashboard Publishing**

After successful development, the Tableau Dashboard and Story are published on **Tableau Public**, allowing the visualizations to be accessed online.

- **Step 6: Web Application Development**

A **Flask** web application is developed to embed the Tableau Dashboard and Story into a single user-friendly interface. Users can access all visualizations through a web browser without opening Tableau Desktop.

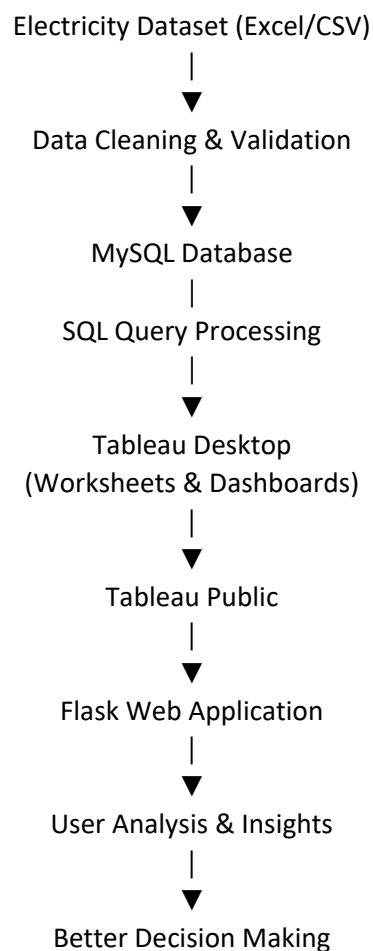
- **Step 7: User Interaction and Analysis**

Energy analysts, government organizations, researchers, and business users interact with the dashboard using filters, maps, and charts to analyze electricity consumption patterns, compare regions, identify trends, and generate business insights.

- **Step 8: Decision Making**

Based on the insights generated by the dashboard, users can support electricity resource planning, improve energy management, identify high-demand areas, and make informed data-driven decisions.

Workflow Diagram



Conclusion

The proposed **Electricity Consumption Analysis** system provides an efficient and user-friendly solution for analyzing electricity consumption data. By integrating **MySQL**, **Tableau**, and **Flask**, the system delivers interactive dashboards, meaningful visualizations, and business insights that support better electricity management and data-driven decision-making. It also provides a scalable foundation for future enhancements and advanced analytics.